



China and AI research

China wants to be at the forefront of AI research by 2030, and has taken the lead in 2019. <https://digit.in/jan20-77>



AI Experiments

Get familiar with machine learning by checking out Google's simple experiments using the tech. <https://digit.in/jan20-78>

an 'electronic brain' or an artificially-intelligent non-living entity.

THE DECADE THAT TRULY STARTED IT ALL

The 1950s saw the concept of AI being catapulted into high gear since many advances in the field of AI came to fruition at this time. The first major success in this decade was when Alan Turing, a famed mathematician, proposed 'The Imitation Game'. According to Turing, a machine that could converse with human beings without them knowing it is the machine would win this 'imitation game' and could be deemed 'intelligent'. This proposal went on to become 'The Turing Test' which became a measure of machine (artificial) intelligence.

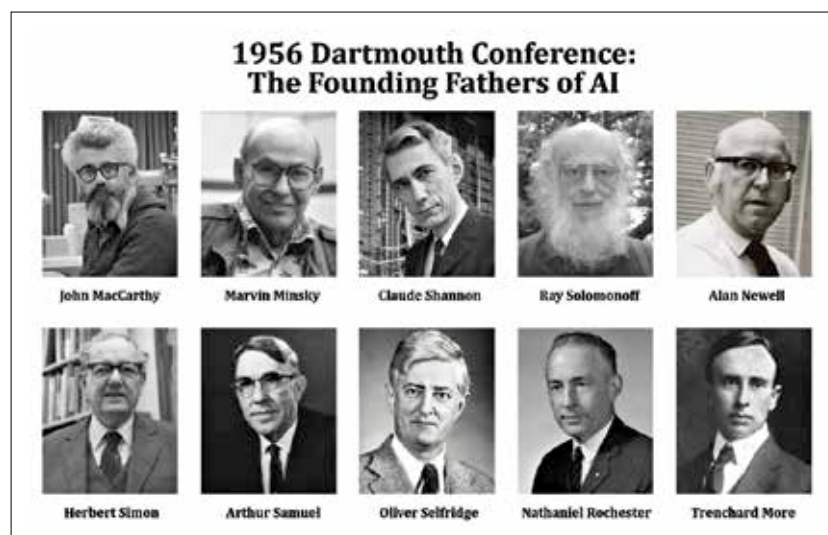
In 1955, Herbert Simon (economist), Allen Newell (researcher) and Cliff Shaw (programmer) conjointly authored 'Logic Theorist', which was the very first AI computer program. Funded by Research and Development (RAND) Corporation, it was designed to imitate the problem-solving skills of humans.

However, the greatest success arguably in this decade was when computer scientist, John McCarthy arranged the Dartmouth Summer Research Project at Dartmouth College in 1956. The term 'Artificial Intelligence' was coined at this conference by McCarthy. 'Logic Theorist', which we mentioned above, was unveiled at the Dartmouth Conference. Following this, several research centres popped up across the US to explore AI and its potential.

In 1958, McCarthy also developed Lisp, which is one of the most popular programming languages for AI research and is still utilised to date. A year later, Arthur Samuel, a computer scientist, coined the term 'machine learning' when he was speaking about building a computer that could play a game of chess much better than the person who programmed it (the computer).

RAPID ADVANCEMENTS

Starting from 1957, all the way through the 1960s, research in the field of AI and innovations around it began flourishing. Computers were faster now and



The scientists that orchestrated the Dartmouth Conference in 1956

could store more information. They also became more easily accessible and cheaper. The innovators in the field of ML created better algorithms and became accustomed to assigning the right algorithm to the right problem. Innovations such as SAINT, a heuristic program for symbol integration in calculus and STUDENT, an AI program that solved algebra word problems became more prevalent due to the influx of capital in this field. Newell and Simon's 'General Problem Solver' and Joseph Weizenbaum's 'ELIZA' programs began exhibiting promise towards the goal of interpreting spoken languages and problem-solving by machines. Such achievements led to government-controlled agencies such as the Defense Advanced Research Projects Agency (DARPA) to begin funding AI research.

The government seemed to be especially interested in innovations surrounding machines that could transcribe and translate spoken languages and those with high throughput data processing. Films such as 2001: A Space Odyssey and Star Wars started featuring sentient machines, HAL and C3PO respectively, in these movies, which were largely based on the concept of AI. In 1970, Marvin Minsky, an American cognitive scientist, confidently stated to Life Magazine, "(In) three to eight years we will have a machine with the general intelligence of an average human being".

However, even though the basic proof of concept was present, AI still had a long way to go before the processes of natural language processing (NLP), abstract thinking and self-recognition in machines could be perfected, or even usable on a mass level.

OBSTACLES LEADING TO 'AI WINTERS'

Once the feverishness around AI started clearing, the obstacles in this field began revealing themselves. Computer scientists were finding it increasingly difficult to inject human-like intelligence in machines. The most prominent obstacle was the lack of computational power. Computers, at the time (around 1980s), weren't technologically-advanced enough to store and process large volumes of data required to bring AI applications such as vision learning to fruition. As governments' and corporations' patience began dwindling, so did the funding. Therefore, from the mid 70s to the mid 90s, there was an acute shortage of funding for researching AI. These years were termed the 'AI Winters'.

The AI project in Britain was partially revived in the 1980s by the government to compete with Japanese AI innovation in the field of expert systems, which mimicked decision making processes of human experts. Additionally, John Hopfield (scientist) and David Rumelhart (psychologist) popularised